Chapter 4

Instruction Format

The format of instructions in the Z8000 architecture is quite regular. The general instruction format uses a two bit field to select the addressing mode, a five or six bit field for the operation code (opcode), a four bit field to select the source operand and a four bit field to select the destination operand.

In the addressing mode field the bit combination 00 usually selects either immediate data or Indirect Register addressing, the bit combination 01 selects either Direct addressing or Index addressing and the bit combination 10 selects Register addressing. The choice between Immediate and Indirect Register or between Direct and Index is made using one of the bit combinations in bits 7-4. This is why R0 cannot be used with Indirect Register or Index addressing.

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte	ad mode		opcode					0	source or destination				source or destination			
Word	ad mode		opcode					1	source or destination				source or destination			
Long	ad mode		opcode						source or destination				source or destination			

General Instruction Format (first word of instruction)

The bit combination 11 in the address mode field is used to specify the compact format for the instruction. Four of the most commonly used instructions have their own compact format.

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LDB	1	1	0	0	destination				byte data							
CALR	1	1	0	1	offset											
JR	1	1	1	0	condition code				offset							
DJNZ	1	1	1	1	register				W	offset						

Special Compact Instruction Format

Some infrequently used or complex instructions require two words to encode all of the information. All EPU instructions also require at least two words.

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	0	0	0	0	register				register				condition code			
	0	0	0	0	register				0	0	0	0	identifier			
	0	0	0	0	register				0	0	0	0	0	0	0	0
EPU	reserved for EPU use												iteration count			
EPU	reserved for EPU use				register				reserved for EPU use				iteration count			

General Instruction Format (second word of instruction)